Decawave Is Now Qorvo

DW3000 Nearby Interaction QSG

Version 1.1

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1 INTRODUCTION

Qorvo DW3000 chip series are approved by Apple® for the purpose of evaluating UWB enabled accessories to leverage Apple's Nearby Interaction framework. Qorvo DWM3000 and DWM3001C modules can interact with Apple® products that include the U1 chip by performing a Two-Way-Ranging with them.

Apple® Nearby Interaction framework with Third Party Accessories has two phases:

- First, is the initial Out Of Band discovery (BLE/WiFi/etc). In this case BLE is used to exchange configuration settings between U1 enabled device and an Accessory device. During this phase, the Accessory is sending the preferred configuration data (Accessory Configuration Data) to the Apple® device. Apple® unit provides its configuration of choice back to the Accessory (Apple Shareble Configuration Data).
- After exchanging of settings, the U1 enabled device starts the UWB session and runs the Two-Way-Ranging between U1 chip and Accessory.

The Two-Way-Ranging protocol used in Apple® Nearby Interaction with Third Party is powered by FiRa© standard. In this standard either side shall be configured as a Controller+Initator and another side shall be configured as a Controlee+Responder. The Accessory can be pre-configured for a specific role.

The Apple® approved Nearby Interaction Qorvo Development Kits are equipped with Nordic BLE SoCs, which provide the Out Of Band discovery over BLE, while the communication with U1 enabled device is provided with the Qorvo DW3000 series chip.

Qorvo offer a UWB module variant (DWM3001CDK) or an Arduino™ compatible Shield (DWM3000EVB), which can be connected to the Nordic Development Kit. Supported Nordic Development kits:

- nRF52-DK
- nRF52833-DK
- nRF52840-DK

This document describes how to get started with the Qorvo solutions for Apple® Nearby Interaction.

2 QORVO UWB PRODUCTS AND NORDIC DK

UWB solution of DWM3000 and DWM3001C integrated modules allows the exploration of Nearby Interaction out of box and using these modules in the end-product designs. For development purposes DWM3000 supplied on the DWM3000-EVB, which is an Arduino™ extension, suitable for a range of Nordic Development Kits.

2.1 DWM3000EVB Shield

The DWM3000EVB (Figure 1) is an Arduino™ form-factor compatible shield designed for the evaluation of the DWM3000 UWB module, which is composed by a DW3110 UWB IC and a Ceramic UWB antenna.

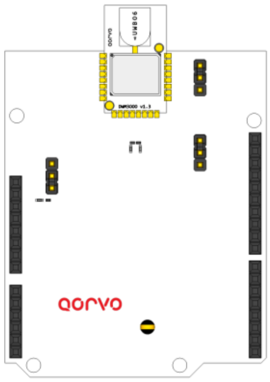

Figure 1 - DWM3000EVB Shield

As shown in the Figure 2 the DWM3000EVB Shield is attached to the Nordic DK's Arduino™ Interface to provide the UWB functionalities.

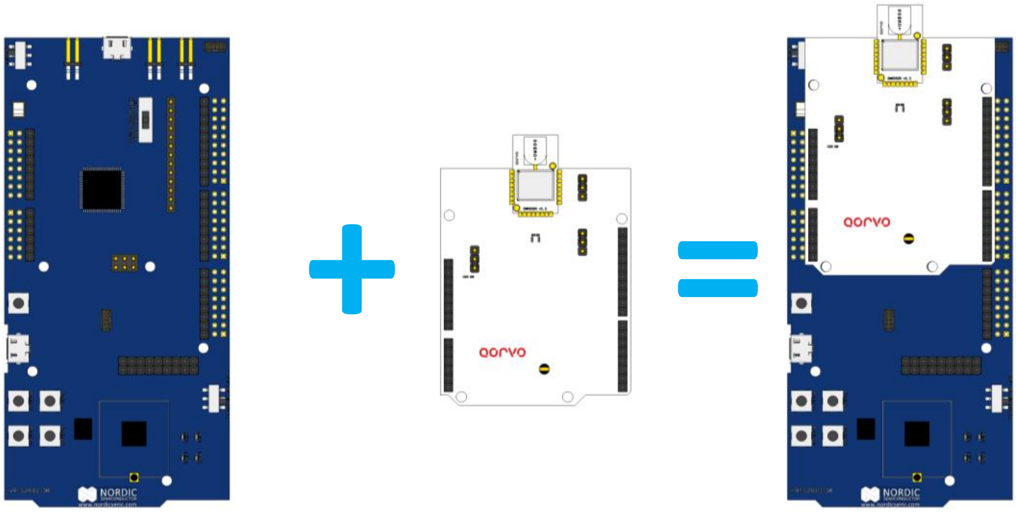

Figure 2 - DWM3000EVB Arduino and a nRF52840/833-DK

2.2 DWM3001CDK Design Kit

The DWM3001CDK (Figure 3) is a Qorvo Design Kit designed for the evaluation of the fully integrated DWM3001C UWB module, supporting a DW3110 UWB IC, PCB UWB antenna, accelerometer and powered by a nRF52833 BLE SoC.

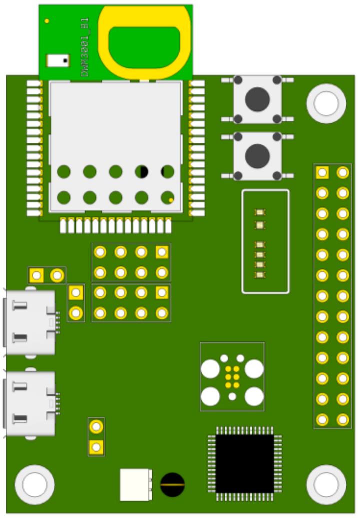

Figure 3 - DWM3001C Design Kit

The DWM3001CDK has an on-board J-Link for flash and debugging via USB (J9), and a direct connection to the DWM3001C USB Interface (J20).

2.3 Applying power to the Nordic DKs

The Nordic nRF52840-DK and nRF52833-DK boards have similar layouts, they are divided into two parts: A J-LINK part, located on the upper part of the PCB, and the target nRF52840/833 microcontroller. To power the board, connect the Nordic nRF52840/833-DK board to a USB power supply with a USB 'Type-A to Micro-B' cable attached to the USB connector marked J2.

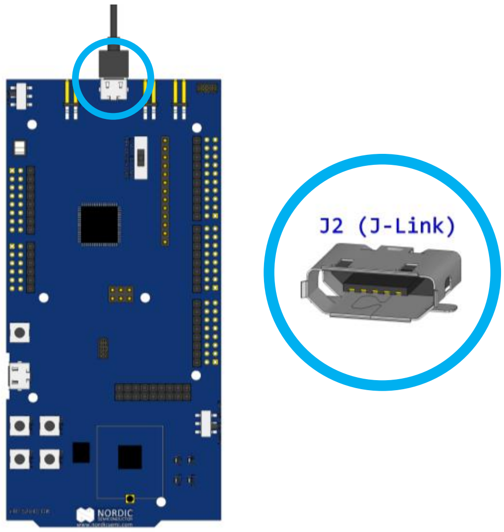

Figure 4 - Powering the boards (J2 USB connector)

The nRF52840/833-DK has two switches that define the power supply for the board (Figure 5):

- SW9 (up in the middle): Switches the power supply between Li-PO (J6), VDD (from J-Link) or USB (from nRF USB).
- SW8 (top left): ON/OFF Switch.

To work from J2 (J-Link) power, SW9 must be switched to VDD (middle position) and SW8 should be in the position ON.

Note
For more details on the power modes, check the User Manual of your Nordic Development Kit

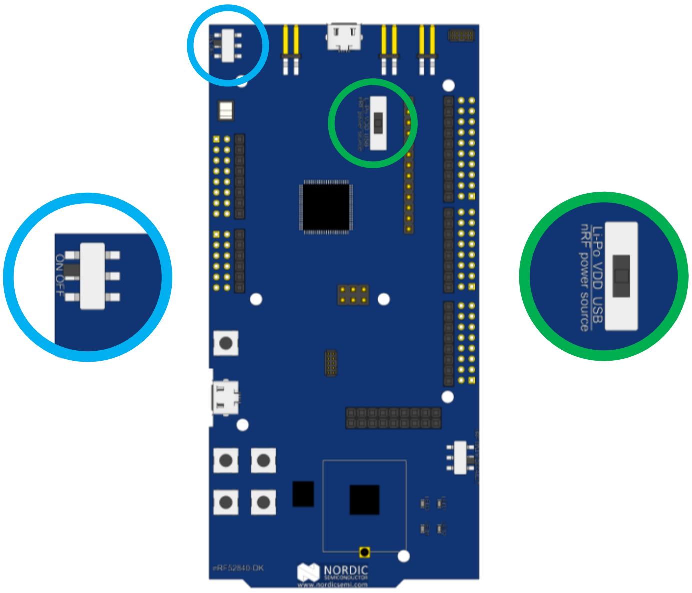

Figure 5 - SW8 (blue) and SW9 (green) position

2.4 Applying power to the DWM3001CDK

Like the Nordic DKs, the DWM3001CDK has an on-board J-Link and a User USB. The board can be powered by both USBs (J20 or J9), or via the RPI interface (from the Raspberry Pi compatible IO) and/or the VBAT connector.

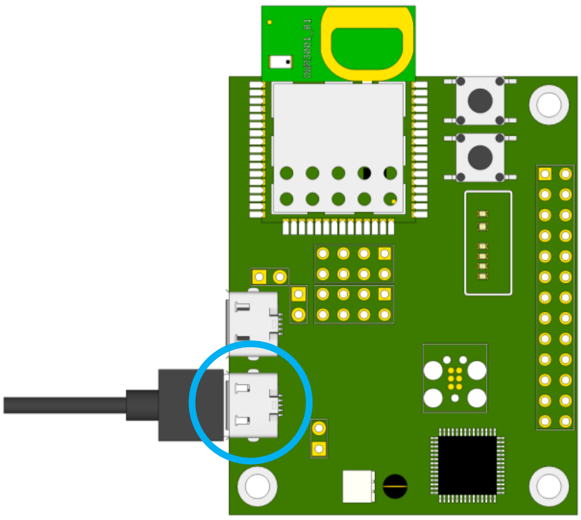

Figure 6 - Powering the DWM3001CDK (J9 USB connector - J-Link OB)

By default, the power supply for the DWM3001CDK is the J9 USB connector, which is also the J-Link OB USB connection used for programming and debug.

3 PROGRAMMING/UPGRADING FIRMWARE

The Nordic nRF52840/833/832-DKs and the DWM3001CDK have an on-board SEGGER J-LINK debugger, allowing to program and debug the target microcontroller. The J-LINK should be connected to a PC, to program the Nearby Interaction example firmware in the boards.

3.1 Application and Drivers

The application to program the Nordic SoCs is the "J-Link Software and Documentation pack", it is available free of charge for any SEGGER J-LINK devices at:

https://www.segger.com/downloads/jlink/

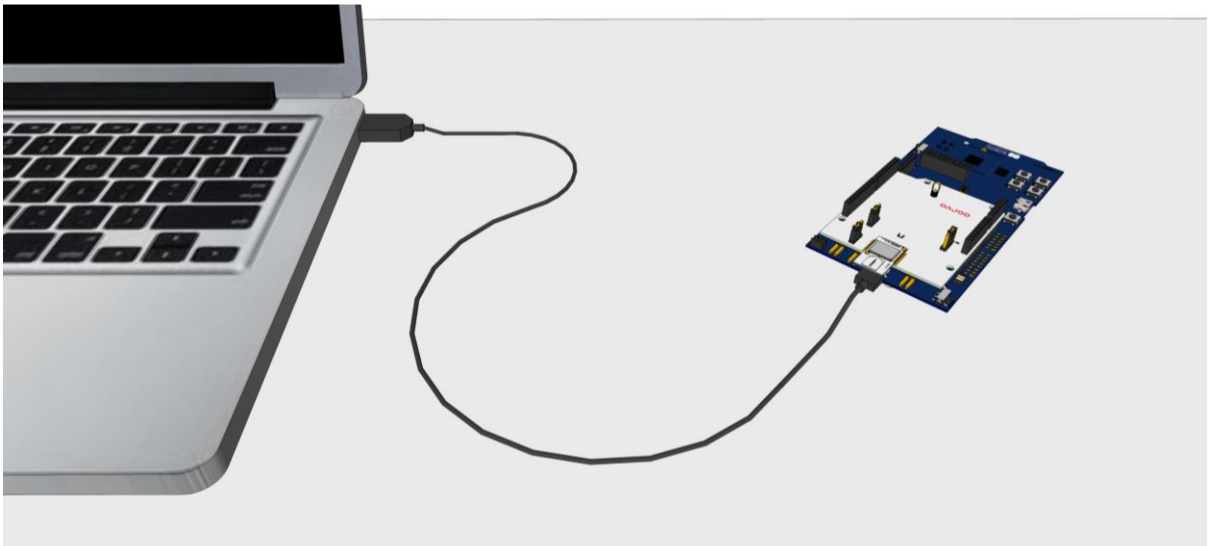

Figure 7 - Connect J-Link (J2) to PC

The Nordic DK, JTAG-OB and DWM3000EVB shield are by default powered from J2 (J-Link) connection[2,3]. Connecting J2 to the PC will provide both, power to nRF52840/833/832 SoC and connectivity for the J-Link.

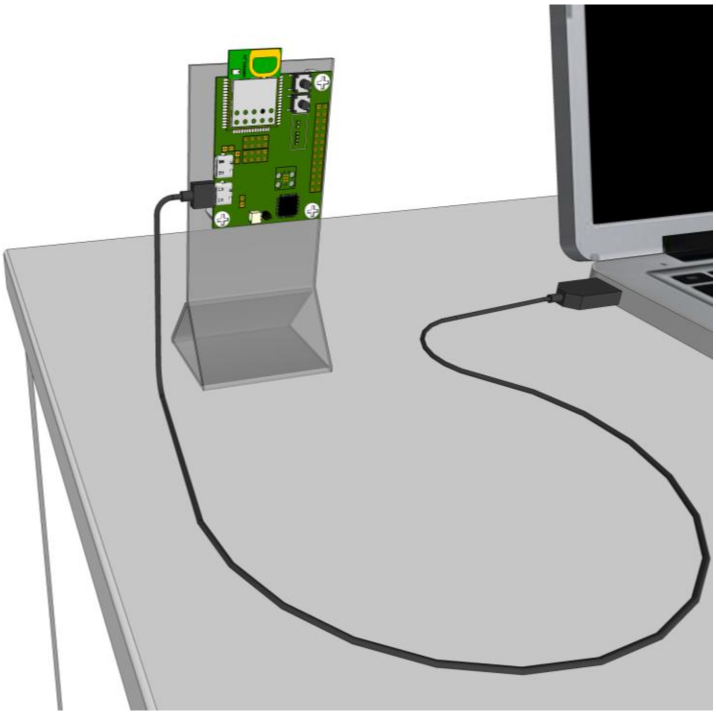

Figure 8 - Connect J-Link (J9) to PC, it will also power the DWM3001CDK

In the DWM3001CDK, the J-Link is connected to the USB (J9) connection, the board can be powered from there[2.4] and it is the only connection required.

3.2 Programming the boards

To program the Nordic nRF52840/833/832 SoCs with the Nearby Interaction firmware provided by Qorvo, the board must be powered and the J-LINK section needs to be connected to a PC using the USB connector (Figure 7 and Figure 8). This will enable the programming of the development board. Open the SEGGER J-Flash Lite application, it will detect a J-LINK connected to the PC. Check the device selected, for the nRF5-DK select "NRF52832_xxAA" device, for the nRF52840-DK select "NRF52840_xxAA" device, for nRF52833-DK and DWM3001CDK select "NRF52833_xxAA" by clicking on the "three dots" button (Figure 9)

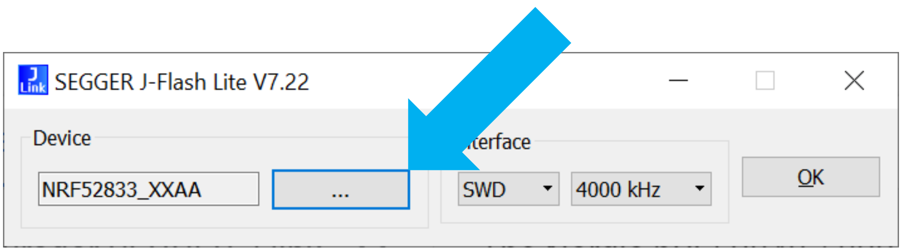

Figure 9 - SEGGER J-Flash Lite, first screen

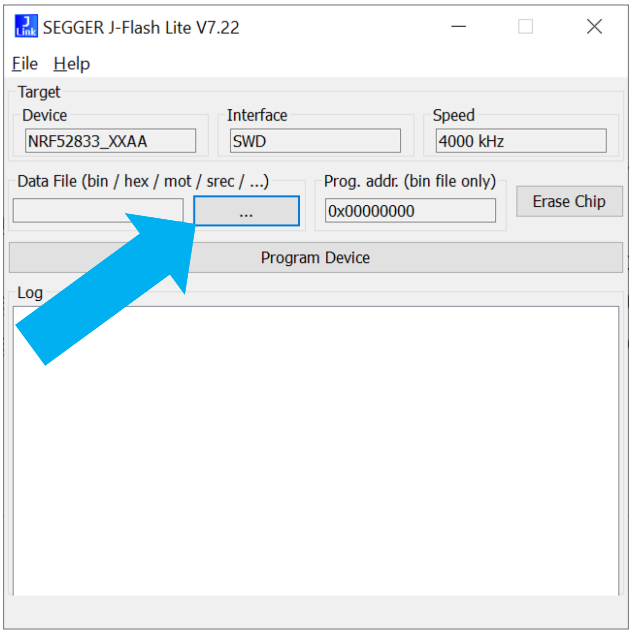

Figure 10 - Three dots button to select the Nearby Interaction example binary

On the new screen click on the three dots button to load the binary file to be programmed to the board. Select the binary file for the correct target pre-built by Qorvo and then click on the "Program Device" button.

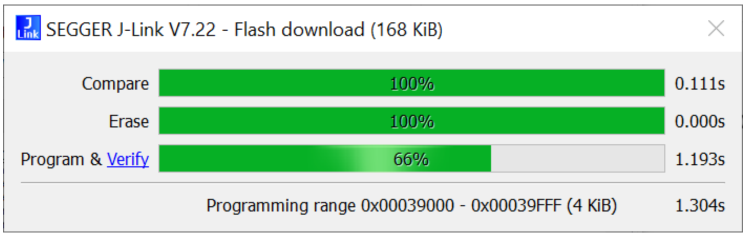

Figure 11 - Programming progress window

The programming process takes a few seconds to finish.

Note
If a 'programming' error occurs, try to 'erase' the chip first ("Erase Chip" button) and then try again to program the board as described in the previous section.

4 NEARBY INTERACTION DEMO APPLICATION

Nearby Interaction with Third Party Accessories was introduced starting from iOS-15. It is required to update an iPhone to the iOS-15 version.

To test the Nearby Interaction Demo, first download the Xcode Beta from Apple website (Applications tab):
[Apple Beta Software Downloads: Xcode 13 Beta]

> **Note**
> Xcode is available only for macOS devices.
> An Apple Developer account is required to download Beta versions.

Then, download and uncompress the Sample Code from:
[Implementing Spatial Interactions with Third-Party Accessories]

Open Xcode, click on File→Open..., navigate to the "ImplementingSpatialInteractionsWithThirdPartyAccessories" folder and open the NINearbyAccessorySample.xcodeproj file.

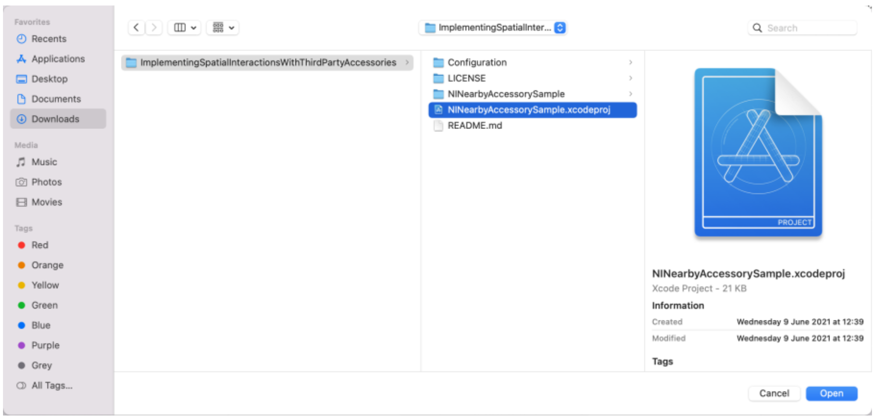

Figure 12 - Open the Sample Project in Xcode 13 Beta

Connect the target iPhone to a free USB port of the macOS device. Compatible devices equipped with a U1 chip are:

- iPhone 11 (11, Pro and Pro Max)
- iPhone 12 (12, Mini, Pro and Pro Max)

> **Note**
> The iPhone must be updated to the iOS 15 Beta.

Select the device (iPhone) on the top navigation bar (Figure 13) and click on the play icon button on the left (Figure 14). The app will be built and deployed to the iPhone.

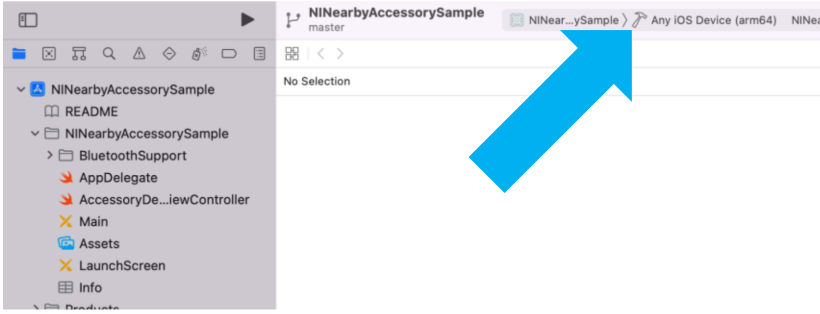

Figure 13 - Change the default iOS device to the connected iPhone model

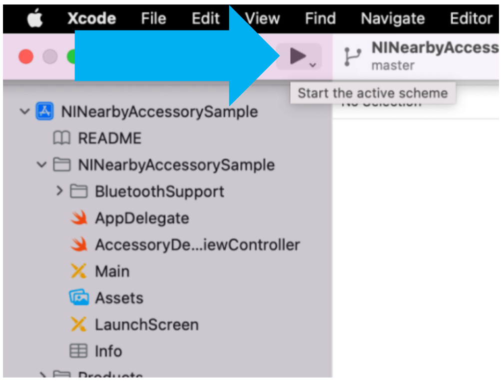

Figure 14 – Build/Start Application in the connected device

When the Sample Application opens it starts to scan for nearby accessories (Figure 15), when a Qorvo device is found it automatically connects and the “Run Session” button is enabled. When clicking on the button, the app will command the Qorvo device to start ranging (Figure 16).

Figure 15 - NI Sample application scanning for Qorvo devices

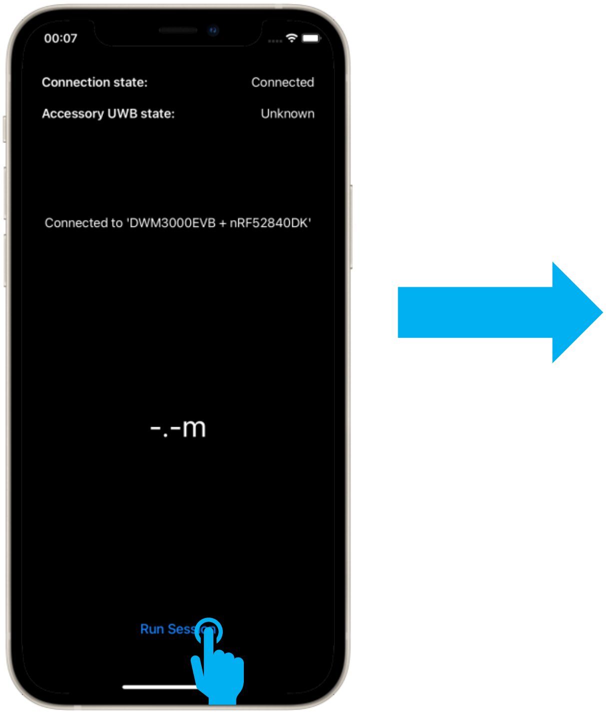

Figure 16 - Connecting to the Qorvo device and starting a NI Session

After every ranging round, new distance information will be available on the screen.

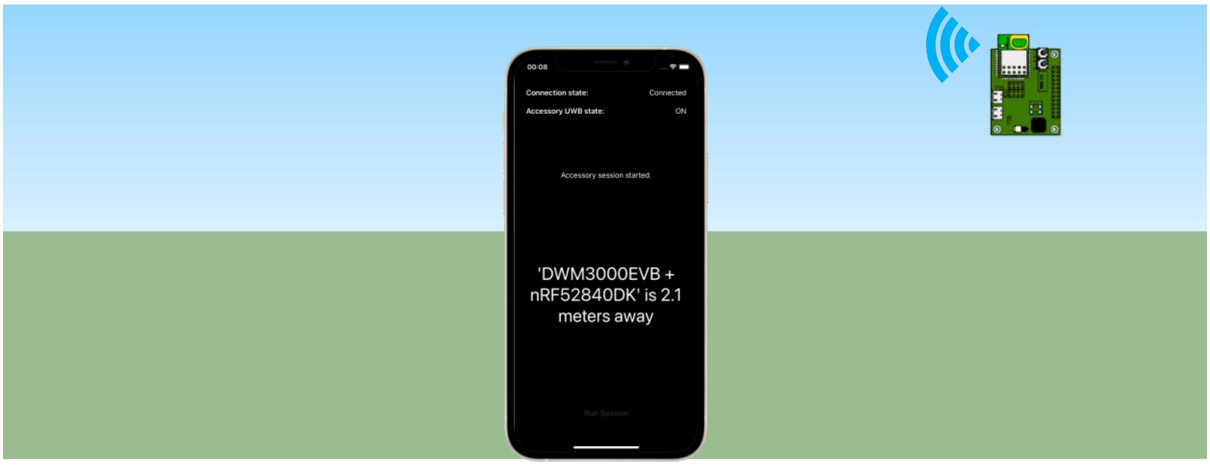

Figure 17 - Real Time positioning information

Moving the Qorvo device towards or away from the iPhone will change the information displayed on the app (Figure 17 and Figure 18).

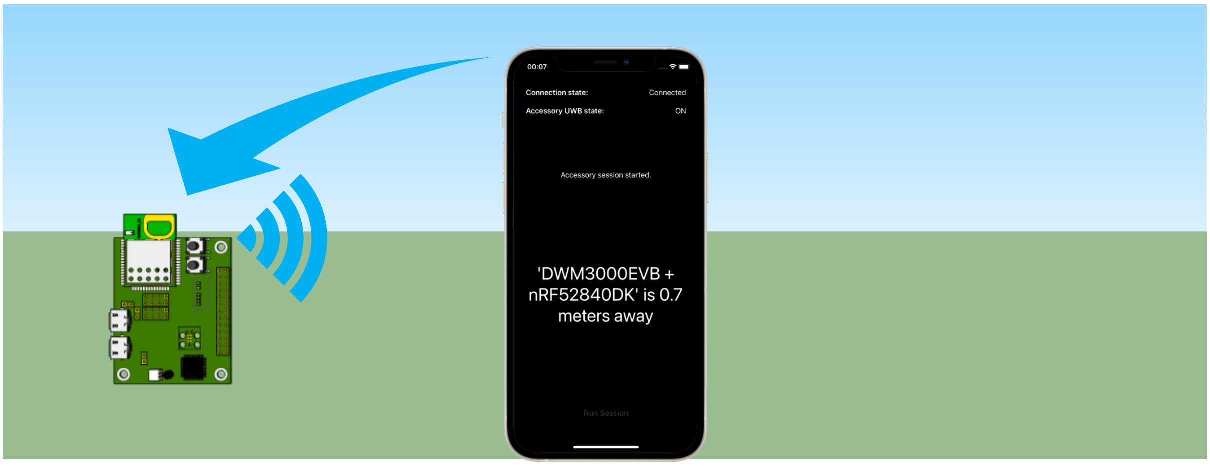

Figure 18 - Change the Qorvo device position to see the positioning info changing

During the Nearby Interaction Demo test, the Qorvo device is sending log information via the J-Link USB. The info can be seen using the “J-Link RTT-Viewer”, included in the “J-Link Software and Documentation pack”**3.1**.

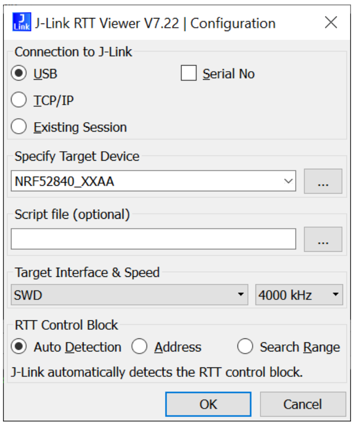

Figure 19 - J-Link RTT Viewer Configuration screen

Connect the board to a PC as shown in 2.3 and 2.4, in the initial screen of J-Link RTT Viewer (Configuration) choose USB, your target device and click on “OK”, the Log information will start (Figure 20).

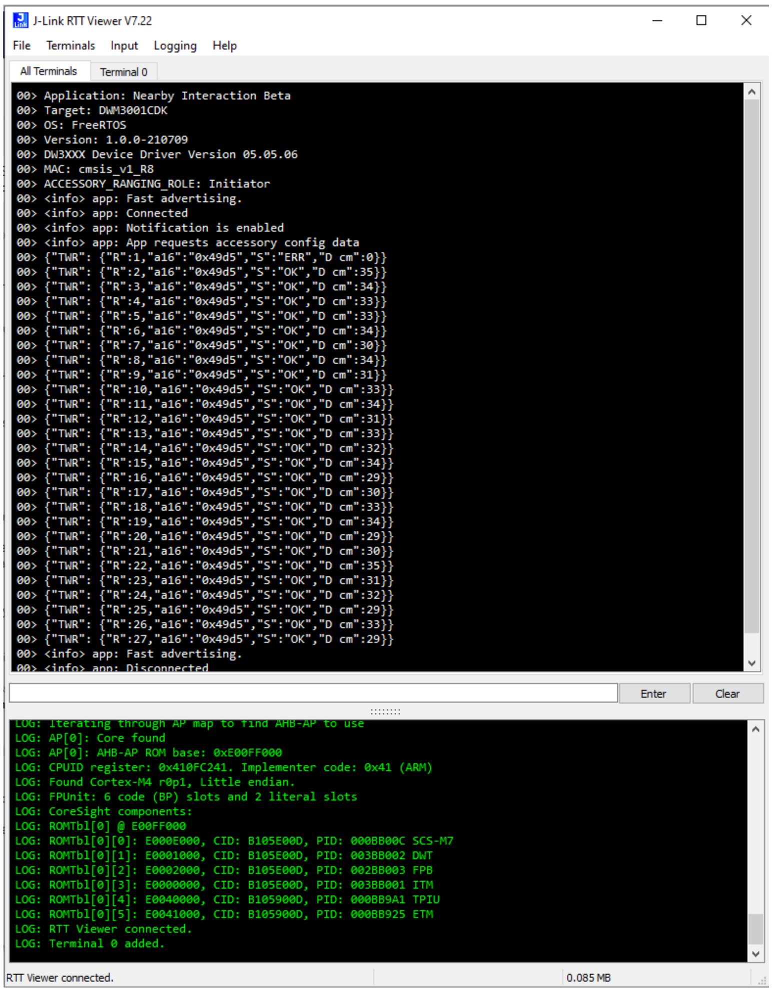

Figure 20 - J-Link RTT Viewer Log Window

5 Developing your own Nearby Interaction Application

To start developing your own iPhone application it is required to register on the Apple® web site as a developer to get an access to the Source code examples of the Nearby Interaction Demo application.

Please visit the following site(s) for more details:
https://developer.apple.com/documentation/nearbyinteraction/implementing_spatial_interactions_with_third-party_accessories

6 Appendix 1 References

1. Qorvo Nearby Interaction resources
   https://www.qorvo.com/feature/uwb-solutions-compatible-with-apple-u1

2. Apple® Nearby Interaction Source code example
   https://developer.apple.com/documentation/nearbyinteraction/implementing_spatial_interactions_with_third-party_accessories

3. Nordic Semiconductors developer's site
   infocenter.nordicsemi.com

REVISION HISTORY

Revision	Date	Description
1.0	04-July-2021	Initial release
1.1	31-Aug-21	Updating links and removing outdated references

7 FURTHER INFORMATION

For further information on this and other Qorvo UWB products, please refer to the website www.qorvo.com/feature/uwb-solutions-compatible-with-apple-u1